

Growth-Scale Rigidity for Surreal and Surcomplex Automorphisms

A negative answer to Kaplan–Krapp–Serra’s Question 5.4,
with convex-valuation and non-lifting theorems

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Abstract

Let F be an ordered field with a total increasing exponential isomorphism $E : (F, +) \rightarrow (F_{>0}, \cdot)$, and let w be any nontrivial convex valuation. We prove that exponential automorphisms stabilizing the valuation ring act faithfully on the value group. More strongly, every nonidentity such automorphism induces a value-group automorphism whose additive displacement is cofinal and coinital. The same conclusion holds after any nontrivial invariant convex coarsening. Applied to the surreal exponential field, this rules out all nonidentity exponential 1-automorphisms, without a strong additivity or coefficient-fixing hypothesis, and answers Question 5.4 in Kaplan–Krapp–Serra, arXiv:2509.22374v3, negatively.

We then exhibit extensive families of ordered value-group automorphisms that have explicit Hahn-field lifts but no exponential lift. These include every nonidentity reweighting confined to a bounded collection of normal-form layers, and every nonidentity positive rational dilation. Finally, for $K = F(i)$ equipped with $L(z) = \log|z|$, we determine its automorphisms and show that the kernel of their action on any invariant nontrivial convex value quotient is exactly complex conjugation together with the identity. All rigidity proofs are algebraic and order-theoretic; none exchanges an automorphism with an infinite sum.

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1 Introduction and the question being answered

A field automorphism can preserve the leading growth of every element while changing lower-order terms. Exponentiation makes this much harder: it can turn an additive error into a change of multiplicative scale. The purpose of this article is to turn that observation into an exact rigidity theorem, and to show that it determines much more than the first leading coefficient.

Kaplan, Krapp, and Serra ask in [1, Question 5.4] whether the surreal exponential field has a nonidentity 1-automorphism. Their Proposition 5.2 rules this out under strong $\mathbb{R}$ -linearity; the unrestricted question is printed on page 10 of version 3. A 1-automorphism fixes every leading term. We prove the stronger assertion that fixing every valuation value already forces the identity. The argument does not assume strong additivity, fixation of the embedded real field, or representation as the exponential of a derivation.

1.1 The principal results

We use additive notation for valuations. Elements outside the valuation ring have negative values; nonzero elements of its maximal ideal have positive values. For an ordered field F , an *ordered exponential* will always mean an increasing group isomorphism

$$E : (F, +) \longrightarrow (F_{>0}, \cdot).$$

In particular, E is onto and has an increasing inverse, denoted by $\log$. If $w : F^\times \rightarrow \Gamma$ is a convex valuation and an automorphism σ stabilizes its valuation ring, its induced value-group action is denoted by τ_σ , so that

$$w(\sigma y) = \tau_\sigma(w(y)) \quad (y \neq 0).$$

Theorem 1.1 (Main rigidity theorem). *Suppose F has an ordered exponential E , w is nontrivial and convex, and σ is an exponential field automorphism stabilizing O_w . If $\sigma \neq \text{id}$, the image*

$$\{\tau_\sigma(\gamma) - \gamma : \gamma \in \Gamma\}$$

is cofinal and cointial in Γ . Consequently, the action on Γ is faithful. For every proper convex subgroup $\Delta \subsetneq \Gamma$, the subgroup stabilizing Δ acts faithfully on Γ/Δ .

The coarsening conclusion requires that the automorphisms under consideration stabilize the chosen convex subgroup. It does not assert that every exponential automorphism stabilizes every coarsening.

For the surreal field, the natural value group is canonically the ordered additive group $(\mathbf{No}, +, <)$. Write

$$v \left(\sum_\gamma a_\gamma t^\gamma \right) = \min \text{supp}(a), \quad t^\gamma = \omega^{-\gamma}.$$

The resulting consequences include

$$\ker(\text{Aut}_{\text{exp}}(\mathbf{No}) \longrightarrow \text{Aut}(\mathbf{No}, +, <)) = \{\text{id}\}, \quad 1\text{-Aut}_{\text{exp}}(\mathbf{No}) = \{\text{id}\}. \quad (1.1)$$

The second equality is the negative answer to the named question. The first is stronger: it does not require preservation of leading coefficients.

The same theorem yields explicit *non-lifting* statements. At the level of the value group, independently rescale real coefficients at any bounded-above collection of normal-form exponents. This gives an ordered additive automorphism and hence a canonical ordered Hahn-field automorphism. Unless all rescalings are

trivial, however, *no* lift can preserve the surreal exponential. Positive rational dilations of the whole value group also fail to lift exponentially, although their displacements are cofinal. Thus cofinal displacement is necessary but not sufficient for exponential lifting.

For the surcomplex field $\mathbf{No}[i]$, define the everywhere-defined, nonzero-input operation

$$L(z) = \log \sqrt{z\bar{z}} \quad (z \neq 0).$$

It uses only the real surreal logarithm, not a choice of complex logarithm. We prove

$$\mathrm{Aut}_L(\mathbf{No}[i]) \simeq \mathrm{Aut}_{\exp}(\mathbf{No}) \times C_2, \quad (1.2)$$

and the kernel of the action on a nontrivial invariant convex value quotient is exactly $\{\mathrm{id}, \text{conjugation}\}$. Marking i removes this final ambiguity.

1.2 Relation to the repository and novelty scope

The inspected repository documentation already has a substantial analysis program, including contour theory, differential equations, and finite spectral theory [9]. We therefore study the automorphism problem rather than repackage those topics. The reference snapshot is specified in the bibliography; this is an independent article, not an edit to that repository.

The concrete literature comparison is the explicit Question 5.4 above. The general cofinal-displacement theorem, its coarsening consequence, the non-lifting families, and the logarithmic-modulus application are presented here as further results. Targeted searches did not identify these statements in this form. This is not a guarantee of priority: the elementary displacement lemma may be familiar in other language, and a specialist may recognize additional antecedents. The distinction between proving a statement and establishing its historical novelty is important.

2 Conventions and the exponential–valuation bridge

2.1 Convex valuations

Let F be an ordered field. A valuation is a surjective homomorphism $w : F^\times \rightarrow \Gamma$ to an ordered abelian group satisfying

$$w(x + y) \geq \min\{w(x), w(y)\},$$

whenever the expression is defined; set $w(0) = \infty$. Its valuation ring, maximal ideal, and group of units are

$$O_w = \{x : w(x) \geq 0\}, \quad \mathfrak{m}_w = \{x : w(x) > 0\}, \quad O_w^\times = \{x : w(x) = 0\}.$$

Convexity means that O_w is order-convex in F . Equivalently,

$$0 < x \leq y \implies w(x) \geq w(y). \quad (2.1)$$

For completeness, $0 < x/y \leq 1$ belongs to a convex valuation ring, giving one direction. Conversely, (2.1) puts every element between 0 and an element of O_w back in O_w ; signs give convexity. All nonzero rational numbers are units for a convex valuation. Indeed, for $n \geq 1$, the valuation inequality gives $w(n) \geq 0$, while $1/n \in [0, 1]$ gives $w(1/n) \geq 0$.

If $\sigma(O_w) = O_w$, then σ also preserves its units. Consequently the formula

$$\tau_\sigma(w(y)) = w(\sigma(y))$$

is well-defined. It is additive and bijective. Since $w(y) \geq w(z)$ is equivalent to $y/z \in O_w$, it preserves the order. This constructs the induced value-group automorphism without any choice of monomial representatives.

We will distinguish carefully between *stabilizing the valuation ring* and *fixing the values pointwise*. The former allows $\tau_\sigma \neq \text{id}$; the latter says $w(\sigma y) = w(y)$ for every $y \neq 0$. Faithfulness concerns the kernel of this action, not the triviality of the whole automorphism group.

2.2 The logarithmic unit subgroup

Lemma 2.1 (The bridge). *Let E be an ordered exponential on F , and let w be a nontrivial convex valuation. The map*

$$\nu : F \longrightarrow \Gamma, \quad \nu(x) = w(E(x)), \quad (2.2)$$

is a surjective additive, order-reversing map. Its kernel

$$H_w = \{x : E(x) \in O_w^\times\} \quad (2.3)$$

is a proper convex additive subgroup of F .

Proof. The exponential law gives $\nu(x + y) = w(E(x)E(y)) = \nu(x) + \nu(y)$. If $x \leq y$, then $E(x) \leq E(y)$, so convexity of w gives $\nu(x) \geq \nu(y)$. For $\gamma \in \Gamma$, choose $a \neq 0$ with $w(a) = \gamma$. Then $\nu(\log|a|) = \gamma$. This proves surjectivity. The kernel is an additive subgroup. If $h_1 \leq x \leq h_2$ and $\nu(h_1) = \nu(h_2) = 0$, monotonicity forces $\nu(x) = 0$. Nontriviality of Γ and surjectivity imply that the kernel is proper. $\square$

Lemma 2.2 (Proper convex subgroups are bounded). *If H is a proper convex additive subgroup of an ordered field F , there exists $B > 0$ such that $|h| < B$ for every $h \in H$.*

Proof. Choose $B > 0$ outside H . If some $|h| \geq B$, convexity between 0 and $|h| \in H$ would put B in H . $\square$

Here “bounded” means bounded by an element of F , not necessarily by an integer or a real number. For example, all finite surreal numbers are bounded by ω . This elementary distinction is decisive.

2.3 The natural valuation on the surreals

The standard facts needed about surreal numbers are their real-closed ordered field structure, Conway normal forms, and the total increasing exponential extending the real exponential [2, 3]. In normal form,

$$x = \sum_{s \in S} r_s \omega^s, \quad (2.4)$$

where S is a reverse well-ordered set and $r_s \in \mathbb{R}^\times$. For $x \neq 0$,

$$\nu(x) = -\max S.$$

The natural valuation ring is

$$O = \{x \in \mathbf{No} : |x| < n \text{ for some } n \in \mathbb{N}_{>0}\}. \quad (2.5)$$

A positive unit u is characterized by bounds $1/n < u < n$ for a sufficiently large positive integer n .

Proposition 2.3. *For the surreal exponential and natural valuation,*

$$\ker(\nu \circ \exp) = O.$$

Proof. If $-n < x < n$, monotonicity gives $e^{-n} < \exp x < e^n$. Since the real numbers e^n are finite, $\exp x$ and its reciprocal are finite, hence $v(\exp x) = 0$. Conversely, if $\exp x$ is a positive unit, choose an integer $n > 1$ with $n^{-1} < \exp x < n$. Taking logarithms gives $-\log n < x < \log n$, so x is finite. $\square$

No normal-form formula for $\exp$ is required in this proposition. In particular, the proof never replaces the Gonshor exponential by the omega-map. The latter has different algebraic and asymptotic properties.

3 An algebraic displacement-amplification lemma

3.1 The exact identity

Let $\sigma : F \rightarrow F$ be a unital field endomorphism, and put

$$D_\sigma(x) = \sigma(x) - x.$$

The map D_σ is additive, but it is generally *not* a derivation. The correct product identity is

$$D_\sigma(ab) = \sigma(a)D_\sigma(b) + bD_\sigma(a). \quad (3.1)$$

Indeed, the right-hand side expands to $\sigma(a)\sigma(b) - \sigma(a)b + b\sigma(a) - ba$. Keeping $\sigma(a)$ rather than replacing it by a is essential.

Lemma 3.1 (Quantitative displacement amplification). *Let F be an ordered field and $\sigma : F \rightarrow F$ a unital field endomorphism. Suppose $a \in F$ is moved by σ , and set $d = D_\sigma(a) \neq 0$. For every $B > 0$, define*

$$b = \frac{2B(1 + |\sigma(a)|)}{|d|}. \quad (3.2)$$

Then

$$\max\{|D_\sigma(b)|, |D_\sigma(ab)|\} > B. \quad (3.3)$$

Consequently, if $\sigma \neq \text{id}$, its additive displacement image $D_\sigma(F)$ is cofinal and coinital in F .

Proof. If both absolute values in (3.3) were at most B , (3.1) and the triangle inequality would give

$$\begin{aligned} 2B(1 + |\sigma(a)|) &= |bd| \\ &= |D_\sigma(ab) - \sigma(a)D_\sigma(b)| \\ &\leq B + |\sigma(a)|B, \end{aligned}$$

a contradiction in an ordered field. Thus absolute displacements have no upper bound in F . Since $D_\sigma(-x) = -D_\sigma(x)$, both signs occur at every required magnitude. Given $c \in F$, apply the result with $B > |c|$ to obtain a displacement greater than c and another smaller than c . $\square$

Surjectivity and order preservation of σ are not needed in this lemma. There are only two probe arguments, b and ab , and all expressions are finite field expressions. In particular, the proof works for self-embeddings and for class-sized fields without a transfinite construction.

Corollary 3.2 (Bounded-displacement rigidity). *A unital field endomorphism of an ordered field whose displacement image is contained in a proper convex additive subgroup is the identity. More generally, a uniform field-valued bound on all absolute displacements forces the identity.*

Proof. Use Lemma 2.2 and Lemma 3.1. $\square$

3.2 A valued-field form of the same principle

The ordered-field proof is especially transparent. A second formulation shows that the underlying effect is not specifically tied to an ordering.

Proposition 3.3. *Let (F, w) be a nontrivially valued field and let $\sigma : F \rightarrow F$ be a nonidentity field endomorphism. The set*

$$\{w(\sigma(x) - x) : \sigma(x) \neq x\}$$

is cointial in the value group.

Proof. Choose a with $d = \sigma(a) - a \neq 0$; then $a \neq 0$ and $\sigma(a) \neq 0$. Given γ in the value group, choose $b \neq 0$ so that

$$w(b) + w(d) < \gamma + \min\{0, w(\sigma(a))\}.$$

Such a b exists because a nonzero ordered abelian group has no least element and w is onto. If both $w(D_\sigma(b))$ and $w(D_\sigma(ab))$ were at least γ , the valuation inequality applied to $bd = D_\sigma(ab) - \sigma(a)D_\sigma(b)$ would give the reverse weak inequality. Thus one of these two displacement values is less than γ . $\square$

The proposition neither assumes nor concludes that σ preserves the valuation. It concerns the size of the additive error, not the induced value-group action. Exponentiation connects those two different notions.

4 Faithful action on every nontrivial convex value quotient

4.1 Cofinal displacement on the value group

Suppose now that σ is an exponential automorphism stabilizing O_w . With $\nu = w \circ E$, there are exact identities

$$\nu(\sigma x) = \tau_\sigma(\nu(x)), \tag{4.1}$$

$$\nu(D_\sigma(x)) = \tau_\sigma(\nu(x)) - \nu(x). \tag{4.2}$$

The first uses $E(\sigma x) = \sigma(E(x))$; the second also uses additivity of ν . Surjectivity of ν yields the equality of images

$$\nu(D_\sigma(F)) = (\tau_\sigma - \text{id})(\Gamma). \tag{4.3}$$

Theorem 4.1 (Cofinal value displacement). *Under the hypotheses of [Theorem 1.1](#), every nonidentity σ has $(\tau_\sigma - \text{id})(\Gamma)$ cofinal and cointial in Γ .*

Proof. By [Lemma 3.1](#), $D_\sigma(F)$ is cofinal and cointial in F . Fix $\gamma \in \Gamma$ and choose $x \in F$ with $\nu(x) = \gamma$. There are displacements $d_- < x < d_+$. Order reversal gives

$$\nu(d_-) \geq \gamma \geq \nu(d_+).$$

Thus $\nu(D_\sigma(F))$ is cofinal and cointial. Equation (4.3) finishes the proof. To obtain strict inequalities at a given target, first choose a larger, respectively smaller, value in Γ , which has no endpoints because it is nontrivial. $\square$

Corollary 4.2 (Faithfulness). *The homomorphism*

$$\rho_w : \text{Aut}_{\text{exp}}(F; O_w) \longrightarrow \text{Aut}(\Gamma, +, <), \quad \sigma \longmapsto \tau_\sigma,$$

is injective. Here $\text{Aut}_{\text{exp}}(F; O_w)$ denotes the stabilizer of O_w .

Proof. Compatibility with composition follows by applying w to $\sigma_1\sigma_2(y)$ and using surjectivity of w . If $\tau_\sigma = \text{id}$, its displacement image is $\{0\}$, which is not cofinal in a nonzero ordered abelian group. Hence $\sigma = \text{id}$. $\square$

There is also a direct kernel proof. If $\tau_\sigma = \text{id}$, (4.2) puts every $D_\sigma(x)$ in the proper convex subgroup H_w . Then Corollary 3.2 applies. The stronger theorem records what happens away from the kernel as well.

Corollary 4.3 (Self-embedding version). *Let $\sigma : F \rightarrow F$ be a unital field self-embedding commuting with E . If $w(\sigma y) = w(y)$ for all $y \neq 0$, then $\sigma = \text{id}$.*

Proof. For every x , exponential compatibility gives

$$v(\sigma x - x) = w(\sigma(E(x))) - w(E(x)) = 0.$$

Apply Corollary 3.2 to H_w . $\square$

This is a self-embedding statement. It does not claim that two embeddings into an arbitrary larger field coincide merely because they induce the same source value-group map: the bound and the amplification arguments must live in the same ambient field as the hypotheses.

4.2 Coarsenings and uniqueness of lifts

For a convex subgroup $\Delta \subseteq \Gamma$, the ordered quotient Γ/Δ is a value group for the coarsened valuation

$$w_\Delta(y) = w(y) + \Delta.$$

It is nontrivial exactly when $\Delta \neq \Gamma$. The usual correspondence between convex coarsenings and convex subgroups can also be formulated for automorphism-invariant valuations; see [5] for this broader valuation-theoretic setting. The proof below needs only the quotient formula itself.

Theorem 4.4 (Coarsening rigidity). *Let $\Delta \subsetneq \Gamma$ be convex. If an exponential automorphism σ stabilizes O_w , stabilizes Δ through τ_σ , and induces the identity on Γ/Δ , then $\sigma = \text{id}$. Equivalently, the stabilizer of Δ in $\text{Aut}_{\text{exp}}(F; O_w)$ acts faithfully on Γ/Δ .*

Proof. Identity on the quotient says

$$(\tau_\sigma - \text{id})(\Gamma) \subseteq \Delta.$$

A proper convex subgroup of an ordered abelian group cannot be cofinal: any positive element outside it is an upper bound. Therefore Theorem 4.1 excludes $\sigma \neq \text{id}$. Alternatively, apply Corollary 4.2 directly to w_Δ . $\square$

Corollary 4.5 (At most one exponential lift). *Fix a nontrivial convex valuation, or an invariant nontrivial convex coarsening, on F . An ordered automorphism of its value group has at most one lift to an exponential automorphism stabilizing that valuation.*

Proof. If σ_1 and σ_2 induce the same value-group action, $\sigma_2^{-1}\sigma_1$ induces the identity and hence is the identity. $\square$

The corollary is a uniqueness theorem, not an existence theorem. The next two sections show that existence is a genuinely restrictive issue.

4.3 A finite witness at a prescribed scale

Proposition 4.6 (Two exponential probes). *Suppose $\sigma(a) \neq a$ and $\eta > 0$ belongs to Γ . Choose $c > 0$ with $w(c) = -\eta$, and set*

$$B = \log c, \quad b = \frac{2B(1 + |\sigma(a)|)}{|\sigma(a) - a|}.$$

Then $B, b > 0$, and for at least one $y \in \{b, ab\}$,

$$|w(\sigma(E(y))) - w(E(y))| \geq \eta. \quad (4.4)$$

The absolute value in (4.4) is the ordered-group absolute value.

Proof. A positive element with negative convex valuation is greater than 1, so $B > 0$ and $v(B) = w(c) = -\eta$. By Lemma 3.1, one of the two chosen arguments satisfies $|D_\sigma(y)| > B$. If $D_\sigma(y) > B$, order reversal gives $v(D_\sigma(y)) \leq -\eta$; if $D_\sigma(y) < -B$, it gives $v(D_\sigma(y)) \geq \eta$. Use (4.2) and exponential compatibility. $\square$

The result makes the detection mechanism explicit: from one moved element and a desired scale, two field expressions followed by exponentiation suffice to exhibit a change at that scale. It is not a finite global algorithm for deciding equality of arbitrary surreal automorphisms.

5 The surreal conclusion and the named open question

5.1 The complete argument specialized to the surreals

The following theorem isolates the answer so it can be checked without using the cofinal-displacement refinement.

Theorem 5.1 (No invisible surreal exponential automorphisms). *Let σ be a field automorphism of $\mathbf{No}$ satisfying*

$$\sigma(\exp x) = \exp(\sigma x) \quad (x \in \mathbf{No}), \quad v(\sigma y) = v(y) \quad (y \neq 0).$$

Then $\sigma = \text{id}$.

Proof. For every $x \in \mathbf{No}$,

$$\begin{aligned} v(\exp(\sigma x - x)) &= v\left(\frac{\exp(\sigma x)}{\exp x}\right) \\ &= v\left(\frac{\sigma(\exp x)}{\exp x}\right) = 0. \end{aligned}$$

By Proposition 2.3, $\sigma x - x$ is finite for every x . In particular,

$$|\sigma x - x| < \omega \quad \text{for every } x \in \mathbf{No}.$$

If σ moved an element a , Lemma 3.1, with $B = \omega$, would produce a displacement with absolute value greater than ω . This is impossible. $\square$

Corollary 5.2 (Answer to Question 5.4). *Every exponential 1-automorphism of $\mathbf{No}$ is the identity.*

Proof. Fixing the leading term fixes its exponent, hence fixes the natural valuation value of every nonzero element. Apply Theorem 5.1. $\square$

This is the only implication from the definition of a 1-automorphism that the proof uses. Leading coefficients play no role. Nor is there an interchange of σ with a Conway-normal-form sum: σ is applied only to individual field expressions.

5.2 Why the argument is not a claim of total automorphism rigidity

Every field automorphism of a real-closed field is order preserving: positive elements are precisely nonzero squares. An ordered field automorphism also preserves Archimedean comparison, because it preserves inequalities involving every positive integer. It therefore stabilizes the natural valuation ring, but it need not fix the value of each element. Our conclusion says that an exponential automorphism is determined by this possibly nontrivial action on values.

In particular, it is incorrect to replace the hypothesis $v(\sigma y) = v(y)$ in [Theorem 5.1](#) by preservation of comparisons $v(y) < v(z)$. The latter is automatic for ordered field automorphisms. The faithful action is the precise statement, and it leaves the image of that action to be understood.

There is another useful conceptual distinction. Suppose one starts with a field automorphism that changes an element only by an infinitesimal. That single local observation does not contradict [Lemma 3.1](#). The contradiction occurs only when a uniform restriction is imposed on the errors at *all* elements. Multiplication amplifies the error somewhere else; exponentiation makes the amplified error visible in the value group.

5.3 A strong nonlocality consequence

For $\sigma \in \text{Aut}_{\text{exp}}(\mathbf{No})$, write τ_σ for its induced automorphism of the additive group $\mathbf{No}$. If $\sigma \neq \text{id}$, then for every surreal $M > 0$ there are $\gamma_+, \gamma_- \in \mathbf{No}$ such that

$$\tau_\sigma(\gamma_+) - \gamma_+ > M, \quad \tau_\sigma(\gamma_-) - \gamma_- < -M. \quad (5.1)$$

This is [Theorem 4.1](#) with its strict form. The conclusion is stronger than merely moving some leading exponent. It excludes any value-group perturbation confined to a bounded range of additive errors, no matter how large that bound is.

The result does not assert that every individual value has to move, or that the set of moved values is an interval. It is a statement about the cofinality of the displacement image. This distinction matters when constructing and rejecting candidate lifts.

6 Explicit value-group automorphisms with no exponential lift

6.1 Two different normal-form levels

In this section $\Gamma = (\mathbf{No}, +, <)$ is the value group. An element $\gamma \in \Gamma$ itself has a Conway normal form

$$\gamma = \sum_s r_s \omega^s.$$

Separately, a field element $x \in \mathbf{No}$ can be written as

$$x = \sum_\gamma a_\gamma t^\gamma, \quad t^\gamma = \omega^{-\gamma}.$$

The index s labels a layer *inside a value*; the index γ labels a monomial *inside a field element*. A reweighting of the first expansion induces a substitution in the second. Keeping these two levels separate avoids a common notational confusion.

6.2 Reweighting a single layer

Fix $s_0 \in \mathbf{No}$ and $c \in \mathbb{R}_{>0}$ with $c \neq 1$. Let $[\gamma]_{s_0}$ denote the real coefficient of ω^{s_0} in γ . Define

$$T_{s_0,c}(\gamma) = \gamma + (c - 1)[\gamma]_{s_0}\omega^{s_0}. \quad (6.1)$$

Proposition 6.1. *The map $T_{s_0,c}$ is a nonidentity ordered additive automorphism of Γ . Its displacement image is exactly $\mathbb{R}\omega^{s_0}$, so it is not induced by any exponential automorphism of $\mathbf{No}$.*

Proof. Coefficient extraction is additive, and the inverse reweights the same coefficient by $1/c$. The sign of a nonzero surreal is the sign of its largest-exponent coefficient. That coefficient is either unchanged or multiplied by the positive real c . Thus the map preserves positivity and hence order. Since $c - 1 \neq 0$, the coefficient $(c - 1)[\gamma]_{s_0}$ ranges over all real numbers; therefore the displacement image is $\mathbb{R}\omega^{s_0}$. Every member of this image has absolute value less than ω^{s_0+1} . It is not cofinal, contrary to [Theorem 4.1](#) for a nonidentity exponential lift. $\square$

The subgroup $\mathbb{R}\omega^{s_0}$ need not be described as convex. The argument requires only that it is bounded. For $s_0 = 0$ and $c = 2$, this example simply doubles the real constant coefficient of each value while leaving all its other coefficients alone. It is a perfectly valid ordered group symmetry that exponentiation forbids at the field level.

6.3 Arbitrary bounded-layer reweightings

Let $c : \mathbf{No} \rightarrow \mathbb{R}_{>0}$ be a specified class function. Define

$$T_c \left(\sum_s r_s \omega^s \right) = \sum_s c(s) r_s \omega^s. \quad (6.2)$$

Each input has set support, so the output is a legitimate surreal number. No uniform bound on the positive real factors $c(s)$ is assumed.

Theorem 6.2 (Bounded-layer obstruction). *Suppose c is not identically 1 and there exists $u \in \mathbf{No}$ such that*

$$c(s) = 1 \quad \text{whenever } s \geq u.$$

Then T_c is an ordered additive automorphism of Γ , but no exponential automorphism of $\mathbf{No}$ induces T_c on values. In particular, this holds whenever the nontrivially reweighted layers form a nonempty set.

Proof. Additivity, invertibility, and preservation of sign follow as in [Proposition 6.1](#); the inverse uses the weights $c(s)^{-1}$. For any γ , every exponent in $\text{supp}(T_c(\gamma) - \gamma)$ is strictly less than u . If the difference is nonzero, its largest exponent is some $s < u$. A real multiple of ω^s , together with lower terms, has absolute value less than ω^u . Consequently

$$|T_c(\gamma) - \gamma| < \omega^u \quad (\gamma \in \Gamma).$$

The displacement is not cofinal, so [Theorem 4.1](#) rules out an exponential lift.

Every set of surreal numbers has a strict surreal upper bound: for a set S the Conway cut $\{S \cup \{0\} \mid \emptyset\}$ supplies one. Thus a set of changed exponents is covered by the stated hypothesis. $\square$

This last clause uses a specifically surreal feature: no set of surreal exponents is cofinal in $\mathbf{No}$. The bounded-layer theorem itself has the weaker and more transparent hypothesis of a strict upper bound.

6.4 The same maps do lift as ordered field automorphisms

The preceding obstructions are not failures of Hahn substitution. That substitution works exactly as expected.

Proposition 6.3 (Canonical field lift). *Let T be any ordered additive automorphism of $\Gamma = (\mathbf{No}, +, <)$. The formula*

$$\widehat{T}\left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right) = \sum_{\gamma} a_{\gamma} t^{T(\gamma)} \quad (6.3)$$

defines an ordered field automorphism of $\mathbf{No}$, fixing $\mathbb{R}$ pointwise and inducing T on its natural value group. It preserves every summable Hahn family.

Proof. An order isomorphism carries a well-ordered support to a well-ordered support. The class image of an input set is still a set, so the right-hand side is a legitimate normal form. Coefficientwise addition is preserved. For multiplication, the equality

$$T(\gamma + \delta) = T(\gamma) + T(\delta)$$

identifies all contributions to each product coefficient. In a Hahn product only finitely many support pairs contribute to a fixed exponent; transport by $\widehat{T}$ gives a bijection between those finite sets of pairs. Thus multiplication is preserved. The inverse is $\widehat{T}^{-1}$. The leading coefficient is unchanged and T preserves the ordering of exponents, so the field order is preserved. Constants have exponent 0, which T fixes. Finally, a summable Hahn family has well-ordered union of supports and only finitely many contributions at any one exponent. Both properties, and the resulting coefficient sums, are preserved by T . $\square$

The construction is the standard external Hahn-field lift; its role in automorphism decompositions is developed in [4]. For a map in Theorem 6.2, Proposition 6.3 is an explicit field lift, but the non-lifting theorem says more than $\widehat{T} \exp \neq \exp \widehat{T}$: it excludes *every* possible exponential lift with that value action, including non-strongly additive ones and lifts that do not fix the embedded real field.

6.5 A cofinal obstruction: positive rational dilations

The bounded-layer examples might suggest that cofinal displacement is the only restriction. It is not.

Theorem 6.4 (Rational dilations do not lift). *Let $r \in \mathbb{Q}_{>0}$ with $r \neq 1$. The ordered additive automorphism*

$$T_r : \Gamma \longrightarrow \Gamma, \quad T_r(\gamma) = r\gamma,$$

has displacement image equal to all of Γ and has the canonical field lift (6.3). Nevertheless, it has no exponential lift to $\mathbf{No}$.

Proof. Because $r - 1$ is a nonzero rational, multiplication by $r - 1$ is a bijection of the divisible group Γ , so $(T_r - \text{id})(\Gamma) = \Gamma$. Assume that an exponential automorphism σ induces T_r . The additive map $\nu = \nu \circ \exp$ is $\mathbb{Q}$ -linear: divisibility and absence of torsion imply $\nu(qx) = q\nu(x)$ for $q \in \mathbb{Q}$. Hence

$$\nu(\sigma x - rx) = r\nu(x) - r\nu(x) = 0.$$

By Proposition 2.3,

$$\sigma x - rx \in O \quad (x \in \mathbf{No}). \quad (6.4)$$

Choose $x = \omega$ and write

$$\sigma x = rx + e, \quad \sigma(x^2) = rx^2 + f, \quad e, f \in O.$$

Multiplicativity gives

$$(r^2 - r)x^2 + 2rxe + e^2 - f = 0.$$

Dividing by x^2 yields

$$r^2 - r + \frac{2re}{x} + \frac{e^2 - f}{x^2} = 0. \quad (6.5)$$

The last two terms are infinitesimal because e, f are finite and $x = \omega$. A nonzero rational cannot equal an infinitesimal. Thus $r^2 - r = 0$, contrary to $r > 0$ and $r \neq 1$. $\square$

The rational hypothesis avoids assuming that an arbitrary exponential automorphism fixes real constants, and avoids any additional assertion about real linearity of $v \circ \exp$. The proof genuinely excludes all lifts with the indicated value action, not just the canonical monomial substitution. It also supplies a multiplication-based obstruction independent of the bounded-displacement obstruction on the value group.

7 Surcomplex automorphisms and logarithmic modulus

7.1 An intrinsic operation that does not choose a complex logarithm

Let F now be real closed and equipped with an ordered exponential. Put $K = F(i)$, where $i^2 = -1$. The standard algebraic complexification has conjugation

$$\overline{a + ib} = a - ib$$

and the F -valued modulus

$$|a + ib| = \sqrt{a^2 + b^2} \in F_{\geq 0}.$$

The nonnegative square root is unique. Its multiplicativity follows by squaring both sides and using positivity. Define

$$L : K^\times \longrightarrow K, \quad L(z) = \log|z|. \quad (7.1)$$

The range is actually contained in F . We keep the codomain K when regarding L as part of a one-sorted algebraic structure; equivalently, one may work with a relation defining its graph on $K^\times$.

There is no use of $\sin$ or $\cos$ at infinite arguments, no choice of an argument of z , and no complex-logarithm branch. This is compatible with, but logically separate from, work on surreal exponential subfields and surcomplex extensions such as [6]. Our structure is specified by (7.1), rather than by an unspecified “surcomplex exponential.”

Lemma 7.1 (Recovery of the real subfield). *The image of L is exactly F .*

Proof. The inclusion $L(K^\times) \subseteq F$ is immediate from the definition. For every $x \in F$, the positive element $E(x) \in K^\times$ satisfies $L(E(x)) = \log E(x) = x$. $\square$

7.2 Exact classification of the structured automorphisms

Write $\text{Aut}_L(K)$ for field automorphisms Σ satisfying

$$\Sigma(L(z)) = L(\Sigma(z)) \quad (z \neq 0).$$

No requirement to preserve F or conjugation is imposed in this definition. Those properties will follow.

Theorem 7.2 (Logarithmic-modulus classification). *Every $\Sigma \in \text{Aut}_L(K)$ has the unique form*

$$\Sigma(a + ib) = \sigma(a) + \varepsilon i \sigma(b), \quad \varepsilon \in \{1, -1\}, \quad \sigma \in \text{Aut}_{\text{exp}}(F). \quad (7.2)$$

Conversely, every map (7.2) belongs to $\text{Aut}_L(K)$. Consequently,

$$\text{Aut}_L(K) \simeq \text{Aut}_{\text{exp}}(F) \times C_2.$$

Proof. By Lemma 7.1 and commutation with L ,

$$\Sigma(F) = \Sigma(L(K^\times)) = L(\Sigma(K^\times)) = L(K^\times) = F.$$

Thus $\sigma = \Sigma|_F$ is a field automorphism of F . Because F is real closed, positivity is characterized by nonzero squares, so σ preserves order. For $y > 0$ in F , $L(y) = \log y$; consequently

$$\sigma(\log y) = \log(\sigma y).$$

Taking $y = E(x)$ and applying E gives $\sigma(E(x)) = E(\sigma(x))$. Therefore $\sigma \in \text{Aut}_{\text{exp}}(F)$. Since $\Sigma(i)^2 = -1$ and the two roots of $X^2 + 1$ in K are i and $-i$, $\Sigma(i) = \varepsilon i$ for a unique sign. This proves (7.2) and uniqueness.

Conversely, let $\sigma \in \text{Aut}_{\text{exp}}(F)$ and choose ε . Formula (7.2) defines a field automorphism by the quadratic extension construction. Order preservation of σ and uniqueness of positive square roots show that

$$|\Sigma(z)| = \sigma(|z|).$$

Since σ commutes with $\log$ on positive elements,

$$L(\Sigma z) = \log(\sigma|z|) = \sigma(\log|z|) = \Sigma(L(z)).$$

The coefficientwise extensions of exponential automorphisms commute with conjugation. Their intersection with $\{\text{id}, \text{conjugation}\}$ is trivial, which gives the direct rather than a merely semidirect product. $\square$

The theorem recovers the real field from the operation itself. Thus its surcomplex conclusion is not just a classification obtained by *assuming in advance* that every automorphism respects a chosen real axis. The range of L supplies that axis intrinsically.

7.3 Extension of the convex valuation

Let w be a convex valuation on F . Put

$$w_K(z) = w(|z|) \quad (z \neq 0). \quad (7.3)$$

Lemma 7.3. *Formula (7.3) is a valuation extending w , with the same value group. For $z = a + ib$,*

$$w_K(a + ib) = \min\{w(a), w(b)\}, \quad (7.4)$$

where $w(0) = \infty$.

Proof. For $z \neq 0$, let $M = \max\{|a|, |b|\} > 0$. Ordered-field inequalities give

$$M \leq \sqrt{a^2 + b^2} \leq 2M.$$

Since $w(2) = 0$, convexity implies $w(\sqrt{a^2 + b^2}) = w(M) = \min\{w(a), w(b)\}$. Multiplicativity of $|z|$ gives multiplicativity of w_K . For addition, apply the valuation inequality in F to each coordinate and use (7.4). The restriction to F is w , and surjectivity onto the original value group follows already from this restriction. $\square$

No positive-definite real norm or Archimedean compactness is involved. The modulus takes values in F , and w_K is the algebraic valuation it induces.

Theorem 7.4 (Surcomplex valuation kernel). *Suppose w is nontrivial. In the stabilizer of O_{w_K} inside $\text{Aut}_L(K)$, the kernel of the induced action on Γ is*

$$\{\text{id}, \text{conjugation}\}.$$

More generally, for a proper convex subgroup $\Delta \subsetneq \Gamma$, the subgroup also stabilizing Δ has this same kernel on Γ/Δ . Within the subgroup fixing i , both actions are faithful.

Proof. By Theorem 7.2, an automorphism is determined by an exponential automorphism σ of F and a sign. Moreover,

$$w_K(\Sigma z) = w(\sigma(|z|)) = \tau_\sigma(w_K(z)),$$

where stabilization of O_{w_K} implies stabilization of $O_w = F \cap O_{w_K}$. Thus the value-group action is exactly that induced by σ . If it is the identity, Corollary 4.2 gives $\sigma = \text{id}$. The two choices of sign leave only the identity and conjugation, and both indeed fix all values. For a coarsening use Theorem 4.4 instead. Fixing i requires the positive sign. $\square$

Corollary 7.5 (Surcomplex application). *For $K = \mathbf{No}[i]$ and $L(z) = \log|z|$, the value-group action of $\text{Aut}_L(K)$ has kernel exactly $\{\text{id}, \text{conjugation}\}$. The same holds on every invariant nontrivial convex value quotient. An automorphism preserving L and i is uniquely determined by any such action.*

The canonical Hahn lifts from Proposition 6.3 also extend coefficientwise to $\mathbf{No}[i]$ and commute with conjugation. For the forbidden value-group maps in Theorems 6.2 and 6.4, these extensions cannot preserve L . More strongly, no L -preserving surcomplex field automorphism has the forbidden value action: restricting it to $\mathbf{No}$ would contradict the corresponding non-lifting theorem.

8 A differential counterpart

The displacement argument also has a useful infinitesimal analogue. This section is a consequence of the same algebraic mechanism, rather than an independent priority claim about derivation theory. Formal sums and the relation between derivations and automorphisms are treated in [8]; none of that machinery is needed below.

Lemma 8.1. *If $\partial : F \rightarrow F$ is a nonzero derivation of an ordered field, then $\partial(F)$ is cofinal and coinital in F .*

Proof. Choose a with $d = \partial a \neq 0$. Given $B > 0$, put

$$b = \frac{2B(1 + |a|)}{|d|}.$$

If $|\partial b| \leq B$ and $|\partial(ab)| \leq B$, the Leibniz identity $bd = \partial(ab) - a\partial b$ gives

$$2B(1 + |a|) \leq B(1 + |a|),$$

a contradiction. Additivity supplies both signs. $\square$

Theorem 8.2 (No globally nonexpanding exponential derivation). *Let E be an ordered exponential on F , and let w be a nontrivial convex valuation. Suppose a derivation ∂ satisfies*

$$\partial E(x) = E(x)\partial x \quad (x \in F)$$

and

$$w(\partial y) \geq w(y) \quad (y \neq 0), \quad (8.1)$$

with the convention $w(0) = \infty$. Then $\partial = 0$.

Proof. Apply (8.1) to $y = E(x)$. When $\partial x \neq 0$,

$$w(E(x)) + w(\partial x) = w(\partial E(x)) \geq w(E(x)),$$

so $w(\partial x) \geq 0$. The same conclusion holds when $\partial x = 0$. Thus $\partial(F) \subseteq O_w$. A nontrivial valuation has a proper valuation ring, and a convex valuation ring is a proper convex additive subgroup. It is bounded in the field order by Lemma 2.2. This contradicts Lemma 8.1 unless $\partial = 0$. $\square$

Condition (8.1) is global. It is much stronger than a condition imposed only on the valuation ring, such as $\partial(O_w) \subseteq O_w$. The theorem must not be used to rule out surreal exponential derivations merely because they have a small-derivation property on finite elements. The obstruction is the universal inequality for *every* nonzero field element.

9 Foundational scope, assumptions, and proof audit

9.1 Set-sized exponential fields

The main arguments are not dependent on the proper-class size of **No**. They apply without change to any set-sized ordered exponential field carrying a nontrivial convex valuation, and to exponential/logarithmic subfields of **No** when those operations are closed in the subfield. Work on such subfields includes [6, 7]. An arbitrary fixed Hahn subfield should not be assumed closed under the surreal exponential or logarithm.

The division of labor among the hypotheses is exact. The ordered-field structure gives the triangle inequality and positive scalars used in amplification. The total exponential identifies additive errors with multiplicative ratios. Its surjectivity makes ν onto the whole value group. Convexity makes ν order-reversing. Nontriviality prevents the value group from collapsing to zero. Stabilization supplies a well-defined induced action. No completeness, saturation, differentiability, or infinite-sum preservation is part of these arguments.

9.2 Class-sized fields and quotients

For **No**, an automorphism is itself a class map. Expressions such as $\text{Aut}_{\text{exp}}(\mathbf{No})$ are convenient metamathematical notation for statements about individual class maps, not assertions that the collection of all such maps is a set or an NBG class. This is the same convention used in [1, Remark 2.1]. Every identity and uniqueness statement here can be read as a statement about specified maps and specified elements.

There is a second, subtler size issue. If Δ is a proper-class convex subgroup of the additive group **No**, its cosets may themselves be proper classes. One need not form a class whose members are those cosets. The statement that a map acts identically modulo Δ means simply

$$\tau(\gamma) - \gamma \in \Delta \quad (\gamma \in \mathbf{No}). \quad (9.1)$$

The proof of [Theorem 4.4](#) uses exactly this relational formulation. For set-sized value groups the ordinary quotient is literal.

For surreal value groups one can, in addition, realize the quotient by canonical representatives. Put

$$S_\Delta = \{s \in \mathbf{No} : \omega^s \in \Delta\}.$$

This is an initial segment of the exponent class. Indeed, if $t < s$, then $0 < \omega^t < \omega^s$, so convexity gives membership at t . For a nonzero surreal γ with leading exponent s , Archimedean comparison with ω^s and convexity give

$$\gamma \in \Delta \iff s \in S_\Delta. \quad (9.2)$$

For the forward direction, some integer multiple of $|\gamma|$ exceeds ω^s ; for the reverse direction, an integer multiple of ω^s exceeds $|\gamma|$. Both multiples stay in the subgroup. Thus Δ consists exactly of the numbers whose support is contained in S_Δ .

Delete from a normal form all terms whose exponents belong to S_Δ . The resulting truncation is a surreal, since its support is a subset of a set support. It is additive, has kernel Δ , and its image is a class of surreal numbers representing the quotient. Nonzero representatives have exponents outside S_Δ and dominate the deleted terms, so their original ordering gives the quotient ordering. This supplies a concrete interpretation of Γ/Δ when desired, without making proper classes into elements.

The same care applies to Hahn lifts: every individual input support is a set, and its image under a specified class automorphism is a set. No proper-class sum is formed anywhere in the construction.

9.3 What has and has not been established

The core claims are mathematical theorems proved above, not conclusions from numerical experiments. Their key dependencies can be summarized without concealing any analytic input:

Result	Essential ingredients
Displacement amplification	Twisted product identity; ordered-field inequality.
Faithful value action	Amplification; the surjective order-reversing map $w \circ E$.
Coarsening rigidity	Cofinal displacement; proper convex subgroups are not cofinal.
Question 5.4	$v(\exp x) = 0$ exactly for finite x ; bounded-displacement rigidity.
Bounded-layer obstruction	Normal forms; leading-exponent domination; cofinal displacement.
Rational-dilation obstruction	Rational linearity of $v \circ \exp$; multiplicativity at ω and ω^2 .
Surcomplex kernel	Range of L ; real closedness; quadratic extension; real-field rigidity.

We have not classified the image of ρ_v , constructed a global surreal composition theory, classified omega-map-preserving automorphisms, or asserted that an arbitrary ordered additive value-group automorphism lifts exponentially. The non-lifting theorems specifically show the last assertion to be false.

The accompanying executable check file tests finite algebraic identities and finite Hahn analogues using exact rational arithmetic. It provides reproducible protection against elementary formula and sign mistakes. It is not a model of the full surreal field and does not certify the theorems involving all elements or class-sized supports. No Lean proof is claimed for this manuscript.

10 Further questions sharpened by the results

The faithfulness theorem replaces a kernel problem by an image problem. The most direct next objective is to characterize the ordered additive automorphisms T of $\mathbf{No}$ that actually occur as τ_σ . The bounded-layer and

rational-dilation examples show that this image is strictly smaller than the full ordered-group automorphism collection, and that a cofinal-displacement test alone cannot characterize it.

Question 10.1 (Image of the value action). Can one describe necessary and sufficient compatibility conditions on $T \in \text{Aut}(\mathbf{No}, +, <)$ for it to admit an exponential lift?

The uniqueness part is already settled by [Corollary 4.5](#). The obstruction in [Theorem 6.4](#) suggests that a useful criterion must retain multiplicative information transported through $\nu = \nu \circ \exp$, rather than only the order structure of the value group. The quotient F/H_w itself is generally not a field: H_w is an additive subgroup, not a field ideal. Thus one cannot simply impose ordinary field multiplication on the quotient and declare the lifting problem solved.

Question 10.2 (Effective obstructions for specified substitutions). For a finitely specified ordered-group substitution T , can one find a finite collection of algebraic/exponential tests that either constructs its unique exponential lift or certifies non-liftability in a chosen exponential subfield?

The two-probe result gives a finite witness *once a candidate nonidentity automorphism is given*; it does not by itself provide the existence test in this question. The distinction between testing a map already defined on the field and lifting a prescribed value-group map must be maintained.

The surcomplex theorem reduces the corresponding image question for L -preserving automorphisms to the real exponential-field question, with only a sign choice for i . This reduction is useful precisely because it needs no global complex logarithm and no analytic path theory.

11 Conclusion

The decisive mechanism is elementary but global. A nonidentity field endomorphism cannot have its additive errors confined to a bounded region of its own ordered field. Exponentiation sends those errors to multiplicative ratios, and a convex valuation reads off their scales. This makes every nontrivial invariant convex value quotient a faithful detector of exponential automorphisms.

For $\mathbf{No}$, it follows that there are no nonidentity exponential 1-automorphisms, without strong-additivity assumptions. The same argument also produces cofinal displacement and exact non-lifting criteria that separate ordered Hahn-field symmetries from exponential symmetries. For $\mathbf{No}[i]$ equipped with logarithmic modulus, the only symmetry left invisible by these value quotients is conjugation. These conclusions are independent of any convergence interpretation of surreal normal forms.

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